Chapter 8

p. 441 7LT Change "4.12" to "4.13".
p. 446 19LB Replace "//" with --".
p. 446 17LB Change "**port** (IN:" to "**port** (Sig_in:".
p. 446 17LB Change two instances of "vector_logic" to "logic_vector".
p. 446 Replace 10LB with the following text:

if IN = k **then** Y(k) <= '1' **else** Y(k) <= '0'.

p. 474 6LB Delete second semicolon at the end of the line.
p. 474 7LB Replace "A: **out**" with "A: **buffer**".
p. 475 19LB Change "**else if** " to "**elsif**".
p. 475 16LB Replace "clock" with "Start".
p. 475 9LB and10LB Replace with the following text:

```
when S_1   => if (A2 = '1') then if (A3 = '1') then next_state <= S_2;
            else next_state <= S_1; end if;
```

p. 475 1LB, 2LB, 3LB Replace "0" with "'0'".
p. 475 20LB Replace 20LB with the following text:

if(reset_b'event and reset_b = '0' then state <= S_idle;

p. 476 13LT Replace "A: **out**" with "A: **buffer**".
p. 476 1LT, 2LT Replace "0" with "'0'".
p. 476 4LT Replace "Nice of you to be running a chaeritycIr_A_F <= '1';" with "clr_A_F <= '1'; **end if**;"
p. 476 5LT Delete the entire line.
p. 476 6LT, 7LT Replace 6LT and 7LT with the following text:

```
when S_1      => incr_A <= '1'; if A2 = '1' then set_E <= '1';
                                 else clr_E <= '1' end if;
```

p. 476 24LT Replace "**then** <= A + 1; end **if**" with "**then** A <= A + "0001"; **end if**".
p. 479 11LT Replace "Design Example" with "Design_Example".
p. 479 17LT, 19LT Replace "std logic" with "std_logic"
p. 479 22LT Replace "sig" with "**end component**;".
p. 479 4LB Replace "<= '1';" with "<= '1' **after** 5 ns;".
p. 479 2LB, 1LB, p. 480 1LT, 2LT, 3LT Replace with the following text:

```
process begin
  t_clock <= '0';
  for k in 0 to 32 loop
        wait for 5 ns;
        count <= count + '1';
        t_clock <= not t_clock;
  end loop;
  wait;
end process;
```

p. 485 14LB, 15LB Replace 14LB, 15LB with the following text:

if clock'event **and** clock '1 **then if** clear = '1' **then** A <= 0;
elsif incr = '1' **then** A <= A + 1;

p. 485 4LB Replace "**if** RST = '0'" with "**if** RST'event **and** RST = '0'"
p. 486 1LB Replace "0" with "'0'".
p. 486 1LT Replace "(Q, Q_not" with (Q: buffer; Q_not"
p. 501 3LB Replace"'001'" with ""001"".
p. 501 2LB Replace"'010'" with ""010"".
p. 501 1LB Replace"'100'" with ""100"".

p. 502 15LB Delete "**begin".**
p. 507 5LB Replace 5LB with the following text:

end component;
begin

p. 508 7LT Replace "'10111'" with ""10111"".
p. 508 8LT Replace "'10011'" with ""10011"".
p. 512 3LT Replace "**if** Q(0)" with "**if** Q(0) = '1'".
p. 523 18LB Replace 18LB with the following text:

generic (R1_size: **integer** := 8; R2_size: **integer** := 4);

p. 523 11LB Add the following text at the end of the line: **end component**;
p. 523 9LB Add the following text to the end of the line: **end component**;
p. 524 1LT Change "controller to "Controller".
p. 524 2LT Replace "Ready," with "Ready **buffer**;".
p. 524 6LT, 7LT Change "**constant"** to "**signal".**
p. 525 3LT Replace 3LT with the following text:
M2: decoder_2x4_df **port map** (D => Decoder_out, A => G1, B => G0, enable => GND);
p. 525 4LT Replace "Ready => Sig1" with "Sig1 => Ready".
p. 525 4LT Replace "Start => Sig2" with "Sig2 => Start".
p. 525 7LT Replace 7LT with the following text:

generic (R1_size: **integer** := 8; R2_size: **integer** := 4);

p. 525 15LT, 16LT, 17LT Replace "**constant"** with "**signal** Std_Logic**".**
p. 525 11LB Replace "CLK, RST, Reset_b" with "CLK, RST_b".
p. 525 6LB Replace 6LB with the following text:

process (R1) **begin**
  Zero <= '1';
  for k **in** 0 **to** R1_size-1 **loop** if R1(k) = '1' **then** Zero <= '0'; **end if**; **end loop**;
end process;

p. 526 5LT replace "integer" with "(**integer**" and replace "8;" with "8);".

p. 526 6LT Replace "R1: o**u**t" with "R1: **buffer**".
p. 526 12LT, 13LT Change "**constant"** to "**signal**".
p. 526 17LT Replace 17LT with
if reset_b'event **and** reset_b = '0' then R1 <= '0'.
p. 526 7LB Replace 7LB with the following text:
if reset_b'event **and** reset_b = '0' then R2 <= '0'.
p. 526 15LB Replace 15LB with the following text:
generic (R2_size: **integer** :=4);

p. 526 11LB Replace "Counter'"with "Counter **is**".
p. 526 4LB Replace "R2 + 1;" with "R2 + '1';".
p. 527 2LT Replace 2LT with the following text:

port (Q: **out** Std_Logic; D, CLK, RST_b: **in** Std_Logic);

p. 527 6LT Replace "**process**" with "**process** (CLK, RST_b) **begin**".
p. 527 7LT Replace 7LT with "**if** RST_b'event **and** RST_b = '0' **then** Q <= '0'"
p. 527 8LT T Replace in two instances "clock" with "CLK".
p. 527 13LT Replace 13LT with the following text:
port (Q: **buffer** Std_Logic; Q_b: out Std_Logic; D, CLK, RST_b: **in** Std_Logic);
p. 527 17LT Replace "**process begin**" with "**process** (CLK, RST_b) **begin**".
p. 527 18LT Replace 18LT with the following text:
if RST_b'event **and** RST_b = '0' **then** Q <= '0';
p. 527 13LB Replace "R2:" with "(R2:".
p. 527 13LB Replace "0);" with "0));".
p. 527 14LB Replace "R1:" with "(R1:".
p. 527 14LB Replace "0);" with "0));".
p. 528 17LT Replace '11111111' with "11111111".p. 528 25LT Replace "//" with "**--**".
p. 535 26LT Replace "precluded" with "precluded in SystemVerilog".
p. 543 In Problem 8.15, replace "*Simulate*" with "Simulate".
p. 547 8LT Replace two instances of "RA == 0" with "RA = '0'".
p. 547 8LT Replace "E := 1" with "E := '1'".
p. 547 8LT Replace "E <= 1" with "E <= '1'".
p. 547 9LT Replace E := 0" with "E := '0'". Replace "E <= 0" with "E <= '0'"

p. 547 14LT In 8.32 change "following always block" to "following **always** block".
p. 551 12LT In 8.44, replace "in3" with "in_3".
p. 551 In 8.45, replace "(Verilog)" with "SystemVerilog".
p. 551 In 8.46, replace "(Verilog)" with "SystemVerilog".
p.551 1LB Delete 1LB
p. 551 In 8.44 replace "**input** in_1, in_2, in3,"
with "(**input** [7:0] in_1, in_2, in_3, **input** clk,".
p. 552 1LT, 2LT, 3LT Delete 1LT, 2LT, 3LT.
p. 552 In 8.47 replace "code" wih "SystemVerilog code".
p. 552 In 8.48, replace "model" with "SystemVerilog model".
p. 553 In 8.51 replace "s2, s, s4" with "s2, s3, s4".

p. 643 15LT Replace "'10_0011'" with ""100011"".
p. 643 16LT Replace "'01_0101" with ""010101"".
p. 643 2LB Replace "'0011'" with ""0011"".
p. 17LT Replace "'01_1010'" with ""011010"".
p. 644 1LB to 9LB Replace all single quotes with double quotes.
p. 645 8LB Replace "'0000'" with ""0000"".
p. 645 7LB Replace "D0 = 1" with "D0 = '1'".
p. 645 6LB Replace "D0 = 0 and D1 = 1" with "D0 = '0' **and** D1 = '1'".
p. 645 5LB Replace "D0 = 0 and D1 = 0 and D2 = 1"
" with "D0 = '0' **and** D1 = '0' **and** D2 = '1'".
p. 645 4LB Replace '0001' with "0001".
p. 649 12LB Replace 12LB with the following text:
if CLK'event **and** CLK = '1' **then** Q <= SI & Q(3:1); **end if**;
p. 649 6LB Add the following text to the end of 6LB: **end component**;
p. 651 In 9LB through 4LB replace "then" with "**then**".
p. 651 8LT Replace "Clock = 1" with "Clock = '1'".
p.651 9LT Replace 9LT with "**if** Load = '1' **then** A <= "0001"".
p. 651 16LB Replace "std_logic_vector (3 **downto** 0)" with "**integer**".
p. 651 11LB Replace "reset = 1" with "reset = '1'".
p. 651 4LB to 9LB Delete quotes.
p. 653 Replace 12LT to 21LT with the following text:

constant s0: Std_Logic_Vector (3 **downto** 0) = "0000";
constant s1: Std_Logic_Vector (3 **downto** 0) = "0001";
constant s2: Std_Logic_Vector (3 **downto** 0) = "0010";
constant s3: Std_Logic_Vector (3 **downto** 0) = "0011";
constant s4: Std_Logic_Vector (3 **downto** 0) = "0100";
constant s5: Std_Logic_Vector (3 **downto** 0) = "0101";
constant s6: Std_Logic_Vector (3 **downto** 0) = "0110";
constant s7: Std_Logic_Vector (3 **downto** 0) = "0111";
constant s8: Std_Logic_Vector (3 **downto** 0) = "1000";
p. 653 19LB Replace "= 1" with "= '1'".
p. 653 15LB Replace "1'b0;: with "'1';".
p. 653 13LB Replace "if (start)" with "if start = '1'".

p. 654 1LT and 2LT: Delete 1LT and 2LT.
p. 658 4LB, 3LB Replace "= 1" with "= '1'".
p. 659 20LT Bold: **when** S1
p. 659 10LB Replace "in std_logic_vector" with "**in** std_logic_vector".
p. 659 Replace 9LB and 8LB with the following text:
clock: **in** std_logic; borrow: **out** std_logic);
p. 659 16LT, 17LT Replace "<= 0" with "<= ''0'".
p. 659 19LT Replace "= 1" with "= '1'".
p. 659 21LT Replace two instances of "= 1" with "= '1'".
p. 664 18LB Replace 18LB with

```
                    constant        s0: Std_Logic := '0';
                    constant        s1: Std_Logic := '1';
```

664 15LB Replace "= 0" with "= '0'".
p. 664 14LB Replace "= 1" with "= '1'".
p. 664 9LB, 10LB Replace "<= 0" with "<= '0'".
p. 664 6LB Replace two instances of "= 1" with "= '1'".
p. 664 5LB, 4LB Replace "= 1" with "'
p. 664 3LB Replace "<= 0" with "<= '0'".
p. 665 10LT Insert "**end component**" at the end of 10LT.
p. 665 13LT Insert "**end component**: at the end of 13LT.
p. 668 4LB Change "8_8" to "8_9".
p. 669 14LB to 10LB Replace "<= 0" with "<= '0'".
p. 669 17LB Replace "reset_b = 0" with "reset_b = '0''.
p. 669 16LB Replace "clock = 1" with "clock = '1'".
p. 669 9LB Replace "**case** state" with "**case** state **is**".
p. 672 2LT Replace "= 0" with = '0'".
p. 672 3LT Replace "= 1" with = '1'".
p. 674 6LT, 7LT Replace "<= 0" with "<= '0'".
p. 674 9LT Replace two instances of "= 1" each with "= '1'".
p. 674 22LT After 22LT insert new line: **end component**;
p. 674 24LT After 24 LT insert new line: **end component**;